

JAMEY McELVEEN

PRINCIPAL SYSTEMS ARCHITECT · WILEY AUTHOR

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EXECUTIVE SUMMARY

Principal Systems Architect with 30+ years delivering enterprise .NET platforms across SaaS, FinTech, and Healthcare. Designed and sustained scalable multi-tenant systems serving nearly 50,000 organizations; led teams of 20+ engineers through complete platform generational shifts. Expert in AI-Augmented Engineering, leveraging LLM-driven tooling (Cursor/Claude) to accelerate legacy modernization and reduce friction on complex refactoring work. Published Wiley author. Currently architecting an open-source mission clinic EMR (Christ Medical) with offline-first sync for field clinics in low-connectivity environments. Actively mentoring peers and colleagues in AI-augmented development workflows.

CORE COMPETENCIES

Languages — C#, ASP.NET Core / .NET 9, T-SQL, TypeScript, JavaScript, Scala, C++, Java	Frontend — Angular, React, Next.js 15, HTML/CSS, Tailwind	APIs & Protocols — REST, WCF/SOAP, WS-Security, OpenAPI
Architecture — SOA, Event-Driven, OOP, Design Patterns, N-Tier, MDD, Strangler, Offline-First	Data — PostgreSQL, MS SQL, Snowflake, Oracle, Redis	SQL Performance — Covering Indexes, RowVersion, RLS, Resource Governor
Cloud & DevOps — Kubernetes, EKS, Docker, Railway, Vercel, CI/CD	Observability — ELK Stack, Structured Logging, OpenTelemetry	Compliance — HIPAA, PHI Masking, BAA, TLS 1.2+, FinTech Audit
AI Tooling — Cursor, Claude, LLM Workflows, Clean-Room PII Protocol	Integration — Epic EMR, Oracle Cloud, Dotmim Sync, GIS/ESRI	Leadership — R&D Mgmt, Agile, Mentoring, ADRs, Technical Writing

PROFESSIONAL EXPERIENCE

Principal Architect, Pro Bono / Kingdom Work

2024 – Present

Christ Medical Mission System · Healthcare / Missions · github.com/christmedical

- Architected an offline-first sync layer using Dotmim Sync so field laptops operate independently and reconcile with the central hub on reconnect: zero data loss, no paper fallback required.
- Implemented Double Metaphone phonetic patient search via PostgreSQL's fuzzystmatch extension, solving name romanization inconsistency across multilingual mission contexts.
- Unified spiritual and medical outcome tracking in a single data model (spiritual=heard|hope|none), treating the whole person as the patient.
- Enforced CI parity via a single gate running ESLint, Vitest, React Testing Library, and production Next.js build locally and in GitHub Actions.

Senior Solutions Architect

2023 – 2025

SecureGive · FinTech / Donation Processing

- Architected high-concurrency public APIs in .NET Core on Kubernetes (EKS) with elastic scale to absorb "Giving Tuesday" traffic spikes: zero downtime during peak donation surges.
- Contributed across the full product surface: the Scala / Angular / PostgreSQL flagship SPA, a React Native mobile app, and an Electron kiosk platform integrated with Ingenico card readers.
- Created, maintained, and documented Angular components using Storybook across the full frontend component library.
- Designed and implemented the client-facing public API in C# / .NET Core over Snowflake, documented with OpenAPI/Swagger and published via Postman (apidocs.securegive.com).
- Designed anti-fraud heuristics to detect and shut down credit-card validation attacks ("carding"), protecting platform integrity and donor funds.
- Engineered idempotent ETL pipelines for PostgreSQL to Snowflake migrations; pipelines restart mid-stream without duplicating financial records or corrupting the ledger.

- Leveraged Snowflake Time Travel for financial audit trails and year-over-year trend reporting without loading the production transactional database.
- Applied FinTech-grade security including immutable audit trails, TLS 1.2+ enforcement, and least-privilege access across all data surfaces.

Senior Software Developer / Modernization Lead

2021 – 2023

McLeod Health · Enterprise Healthcare / HIPAA

- Migrated the entire development organization from file-share-based code storage to Git and GitHub, the organization's first modern source control practice: cleaned 100,000+ commits with BFG Repo-Cleaner, reducing repo size by 80%.
- Implemented CI pipelines for HIPAA-compliant clinical systems, replacing manual deployments with automated, auditable release processes.
- Mentored developers and interns on Git workflows, branching strategy, and pull request culture; socialized changes with leadership to ensure team ownership rather than mandated compliance.
- Introduced structured logging via an in-house ELK stack with a Golden Record PHI scrubbing service, replacing real names and SSNs with synthetic identities in non-production environments. Zero BAA exposure risk.
- Enforced TLS 1.2+ across all WCF and REST endpoints; advocated for data-at-rest encryption and least-privilege access across all clinical data surfaces.
- Managed data integration bridges between .NET applications and Oracle Cloud / Epic EMR platforms serving thousands of clinical staff.

Programming Manager / R&D Architect

1996 – 2021 *

ACS Technologies · Enterprise SaaS · 25 years

* Except May 2005 – May 2006, see Lyle's Data Systems below.

- Strategic architectural lead for a flagship SaaS suite serving nearly 50,000 organizations worldwide; directed a distributed R&D team of 20+ engineers through multiple platform generational shifts.
- Pioneered Model-Driven Development (MDD) for large-scale C# codebases, dramatically reducing boilerplate surface area across the product suite.
- Departed following a strategic leadership transition that fundamentally altered the organization's technical vision and direction.

REALM, MULTI-TENANT SAAS PLATFORM

- Architected multi-tenancy using Row-Level Security (RLS) and SQL Resource Governor to prevent "Noisy Neighbor" degradation, ensuring one large tenant's workload could not starve the other 49,999.
- Built a shard-aware routing layer enabling live zero-downtime tenant migration when organizations exceeded resource thresholds.

FACILITY SCHEDULER, N-TIER SCHEDULING ENGINE

- Designed an N-tier C# / WCF / SOAP architecture with custom WS-Security header processing in the WCF message pipeline, solving cross-network auth challenges the native stack could not handle.
- Eliminated high-concurrency booking conflicts using RowVersion optimistic locking; reduced critical scheduling query overhead by 70% via a covering index eliminating a bookmark lookup.

HEADMASTER, SCHOOL MANAGEMENT ECOSYSTEM

- Modeled non-linear Student-Guardian-Representative hierarchies using M:N relational tables with bitmask permission flags, enabling instant legal-authority validation at 1,000-student scale.
- Architecture maps directly to SC DSS / CAPSS child placement and voucher eligibility workflows: transferable to behavioral health or child-services SaaS domains.

Programming Manager

May 2005 – May 2006

Lyle's Data Systems · Civic Tech / Government SaaS

- Managed three developers delivering Java / Struts MVC government services platforms to municipal clients across the Southeast; personally responsible for requirements gathering, client relationships, and on-site installation.
- Architected a multi-jurisdiction online property tax payment system enabling citizens and banks to pay home, vehicle, and boat taxes online, integrated into each client's existing legacy database infrastructure.

- Independently designed and built "@ Your Request", a citizen issue-reporting platform with photo uploads and real-time progress tracking for municipalities.
- Extended the platform with a contractor bidding module; navigated stakeholder scope misalignment through diagrams and analogies, and a key municipal client overrode internal resistance and funded the full build.
- Performed GIS / ESRI geospatial data conversion migrating legacy systems into MS SQL Server; traveled to Boston for training and gained hands-on experience with ESRI tooling.

COMMUNITY & MENTORING

FIRST Robotics, Technomancers · technomancers.github.io

Technical Mentor, IB Baccalaureate STEM Program

- Mentored high school students in Kotlin software development and circuit/sensor integration for competitive robotics.
- Translated complex engineering concepts into accessible instruction for students with no prior programming background: the same clarity discipline applied to Wiley technical writing and team ADRs.

AI Development, Peer Mentoring

Informal Mentor · Ongoing

- Mentors peers and colleagues in AI-augmented engineering workflows: Cursor, Claude, prompt engineering, and clean-room PII protocols for regulated environments.
- Extensive self-directed study in AI development tooling applied directly to active production projects, not theoretical.

EDUCATION & PUBLICATIONS

B.S. Computer Engineering

Clemson University · 1996

Published Author (Wiley, 2009)

"iPhone Game Development" · Co-authored with Chris Craft

A technical and business guide to creating and selling iPhone games
jameymcelveen.com/book